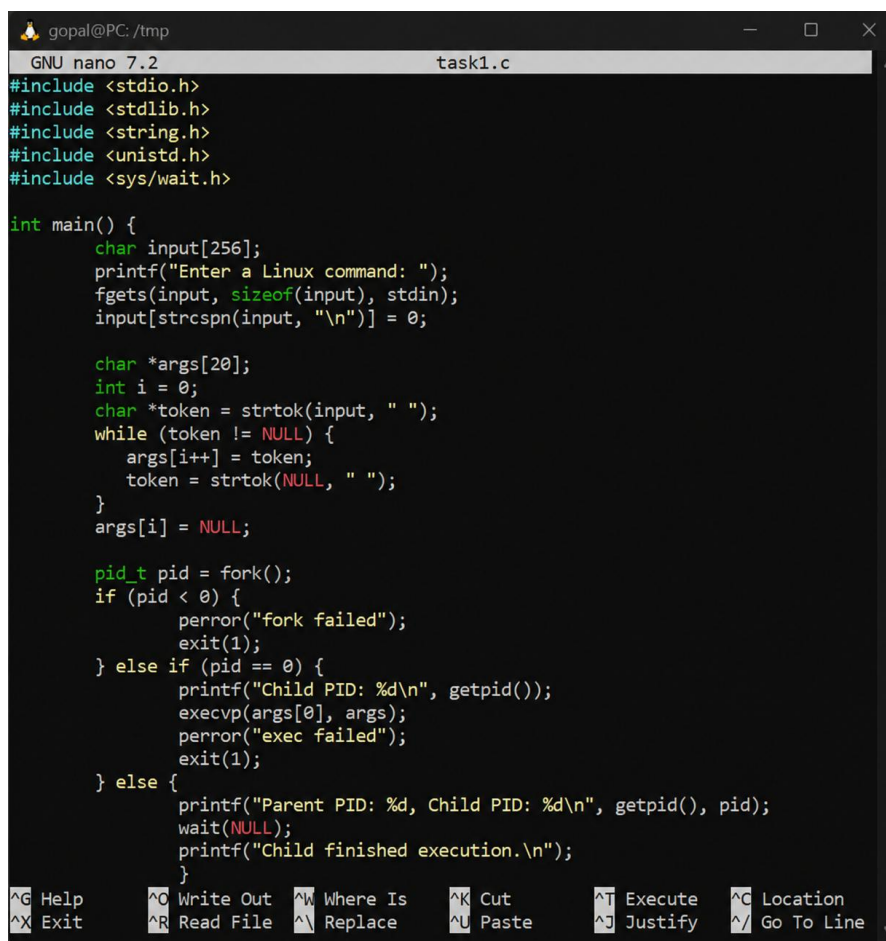


TASK 1

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using `fork()`. 3. Execute the command in the child process using an appropriate `exec()` system call. 4. Allow the parent process to wait for the child using `wait ()`. 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (`uname`, `lscpu`, `lsblk`, `ps`, `top`), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.



```
gopal@PC: /tmp
GNU nano 7.2 task1.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    char input[256];
    printf("Enter a Linux command: ");
    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = 0;

    char *args[20];
    int i = 0;
    char *token = strtok(input, " ");
    while (token != NULL) {
        args[i++] = token;
        token = strtok(NULL, " ");
    }
    args[i] = NULL;

    pid_t pid = fork();
    if (pid < 0) {
        perror("fork failed");
        exit(1);
    } else if (pid == 0) {
        printf("Child PID: %d\n", getpid());
        execvp(args[0], args);
        perror("exec failed");
        exit(1);
    } else {
        printf("Parent PID: %d, Child PID: %d\n", getpid(), pid);
        wait(NULL);
        printf("Child finished execution.\n");
    }
}
```

Help Write Out Where Is Cut Execute Location
Exit Read File Replace Paste Justify Go To Line

OUTPUT :

```
gopal@PC: /tmp
gopal@PC:~$ su gopal
Password:
gopal@PC:/root$ nano task1.c
gopal@PC:/root$ pwd
/root
gopal@PC:/root$ whoami
gopal
gopal@PC:/root$ ls -ld
ls: cannot access '.!': Permission denied
gopal@PC:/root$ cd /tmp
gopal@PC:/tmp$ nano task1.c
gopal@PC:/tmp$ gcc cmd_exec.c -o cmd_exec
cc1: fatal error: cmd_exec.c: No such file or directory
compilation terminated.
gopal@PC:/tmp$ gcc task1.c -o task1
gopal@PC:/tmp$ ./task1
Enter a Linux command: ls -l
Parent PID: 979, Child PID: 982
Child PID: 9822
snap-private-tmp
systemd-private-4bc431fa2fe54f6ebd6553fa9f0b9631-systemd-logind.service-eK1pVF
systemd-private-4bc431fa2fe54f6ebd6553fa9f0b9631-systemd-resolved.service-slbyfB
systemd-private-4bc431fa2fe54f6ebd6553fa9f0b9631-systemd-timesyncd.service-MZohCu
task1
task1.c
Child finished execution.
gopal@PC:~/tmp$ pwd
/tmp
gopal@PC:~/tmp$ date
Wed Aug  5 03:34:48 UTC 2026
gopal@PC:~/tmp$ nano task1
gopal@PC:~/tmp$ nano task1.c
```